

# A GAN-Based Data Augmentation Method for Mitigating Class Imbalance Problem in Histopathological Image Classification

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**Abstract**—This study proposes a Generative Adversarial Network (GAN)-based data augmentation method to tackle the class imbalance issue in histopathological image classification. The proposed GAN generates high-quality images for minority classes, improving data diversity and classification performance. By integrating Global Context Attention (GCA) and UpBlock-CompRes modules, the GAN generates images that closely resemble real data, achieving superior results in FID, IS, PSNR, and SSIM compared to FastGAN. Experimental results demonstrate that the proposed GAN outperforms both the baseline without data augmentation and FastGAN, with noticeable improvements across key metrics. These enhancements highlight the effectiveness of our GAN in generating high-quality images and achieving more balanced classification results.

**Index Terms**—GAN, data augmentation, class imbalance problem, histopathological image classification

## I. INTRODUCTION

Pathological images are the gold standard for diagnosing a wide range of diseases [1], particularly in cancer detection and grading, where their role in clinical diagnosis is indispensable. However, pathological image analysis frequently depends on the subjective judgment of pathologists, making the process time-consuming and prone to human error [2]. In recent years, the rapid advancement of deep learning (DL), particularly in image processing, has demonstrated significant potential in automating pathological image analysis [3]. This technology can enhance diagnostic efficiency and reduce subjective errors, improving both consistency and accuracy in diagnosis [4].

Nevertheless, pathological image classification still faces numerous challenges, with class imbalance being one of the most prominent [5]. In real-world pathological image datasets, certain classes often have significantly more images than others, causing models to perform well on common classes while

overlooking rare ones. This class imbalance may arise from the low incidence of certain diseases or the difficulty in obtaining sufficient image data for specific cases. To address this issue, data augmentation techniques [6] offer an effective solution. To mitigate the impact of class imbalance on model training, GANs [7] are employed to generate new pathological image data, thereby expanding the sample size of minority classes. This approach not only increases sample diversity at the data level but also significantly improves model performance on imbalanced datasets. Through data augmentation strategies, the aim is to provide a more balanced and robust classification solution for pathological image analysis.

This study performs data augmentation on categories with fewer samples, taking the esophageal cancer dataset [8] as an example. The workflow is shown in Fig. 1. Specifically, GAN is employed to generate additional images, increasing the sample size of minority classes. In each round of data augmentation, GAN generates 25 new image samples for each minority class in the original dataset, enhancing both data diversity and quantity, thereby addressing the class imbalance issue. The generated augmented data is then combined with the original dataset to create a more balanced dataset for model training. Subsequently, the Lightweight Receptive Field-based Convolutional Neural Network (LRF-CNN) model is used for classification. The augmented dataset is fed into the LRF-CNN model for multiple rounds of training and testing to evaluate the performance improvements brought by the GAN-based augmentation. The main contribution of this study is the introduction of a GAN-based data augmentation method. Through the design of two modules, GCA and UpBlockCompRes, high-quality histopathological images are generated, which not only increase the sample size of minority classes but also improve the overall balance of the dataset. This enhancement further improves the generalization ability and robustness of the classification model in practical applications.

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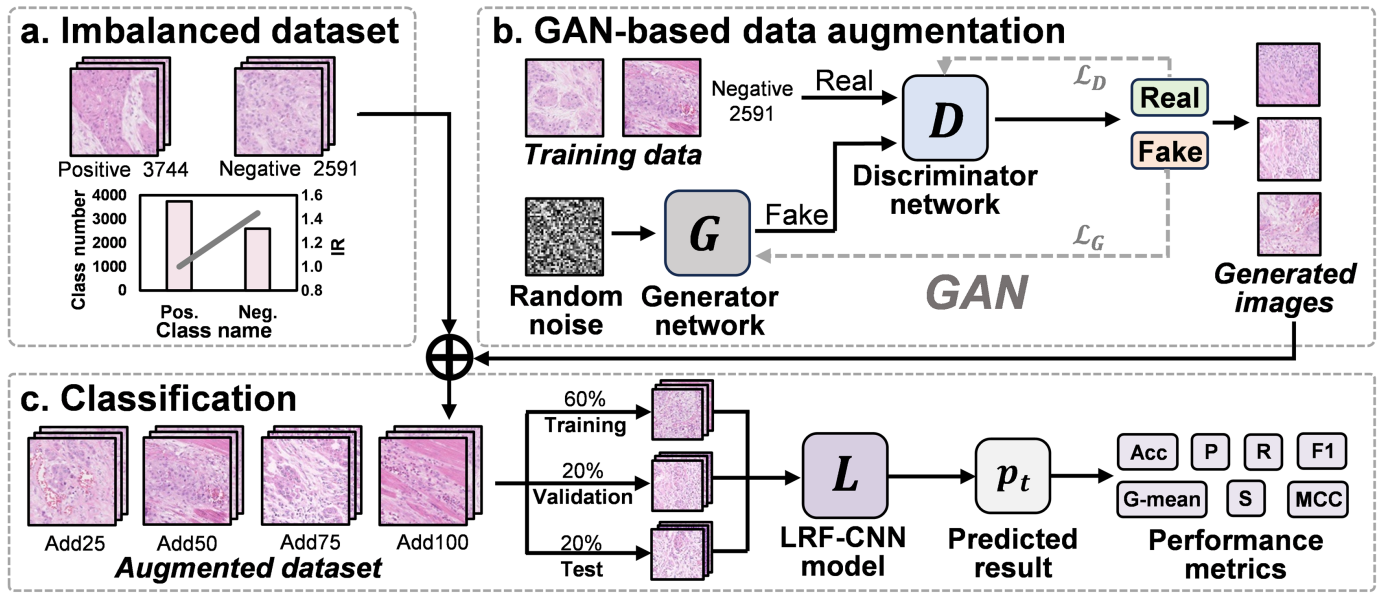


Fig. 1. The workflow for mitigating class imbalance. **a** is the class-imbalanced dataset (esophageal cancer), where the upper part shows examples of two classes, and the lower part shows the class number and the imbalance rate (IR). **b** is data augmentation using GAN. **c** is image classification using LRF-CNN. The numbers in the figure (e.g., 3744) represent the number of a category, while the words (e.g., Positive) represent the name of a category.

## II. RELATED WORK

In recent years, the development of GANs has shown significant potential in addressing class imbalance across various fields, particularly in medical image analysis [9]. GANs effectively generate synthetic data that closely resembles the original data distribution, expanding training sets and mitigating class imbalance. In pathological image analysis, class imbalance often arises due to the low incidence of certain diseases or difficulty in acquiring sufficient image data. Researchers have employed GANs to generate images for minority classes, balancing the dataset distribution [10].

Several studies show that GAN-generated synthetic images can significantly improve model performance on imbalanced datasets. For example, [11] uses GANs to generate breast cancer histopathological images, increasing dataset diversity and enhancing classification accuracy. [12] introduces an innovative GAN to generate minority-class skin lesion images, focusing on improving GAN training stability and image quality. Their approach addresses issues like mode collapse and artifact generation, greatly improving the model's ability to recognize minority classes. [13] applies Wasserstein GAN (WGAN) oversampling to tackle data imbalance in medical image analysis by estimating the distribution of minority categories and generating realistic samples. [14] employs Deep Convolution GAN (DCGAN) to generate normal chest X-ray images, balancing the excess of pneumonia cases and improving CNN classification performance. [15] leverages conditional GAN (cGAN) to generate label-preserving synthetic data by conditioning both the generator and discriminator with additional information, effectively addressing class imbalance in exudate detection for diabetic retinopathy.

Despite the promising applications of GANs in pathological

image data augmentation, several challenges remain in practice [16]. The generation of high-quality images that retain medical authenticity presents a substantial technical challenge [17]. Although GAN-generated images increase data diversity, they may also introduce noise or artifacts that negatively affect model training and generalization [18]. To address these challenges, recent studies have focused on improving GAN design and architecture to balance image quality with the need for diverse data [19]. For instance, [20] incorporates a self-supervised learning mechanism into GANs, enhancing image fidelity while maintaining robustness across different classes, yielding promising results in medical image generation. [21] proposes an extended GAN model incorporating a variational autoencoder and an external memory mechanism, enabling the generation of synthetic data that accurately reflects imbalanced class distributions in clinical variables.

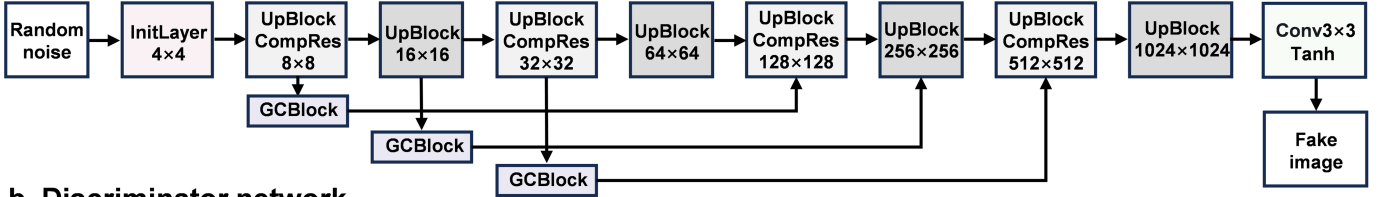
In conclusion, although GANs have shown promising potential in addressing class imbalance in pathological image analysis, challenges such as maintaining medical authenticity and avoiding artifacts persist. Building on this foundation, the GAN proposed in this study introduces key innovations, including a GCA module and an improved UpBlockCompRes upsampling strategy. These enhancements enable the generation of high-quality images from small datasets, effectively mitigating class imbalance while preserving image fidelity.

## III. METHODS

### A. GAN-based Data Augmentation

The proposed GAN generates histopathological images of few-shot classes for data augmentation. Inspired by FastGAN [22], this GAN enhances feature extraction and image generation by integrating a GCA module and an optimized Up-

### a. Generator network



### b. Discriminator network

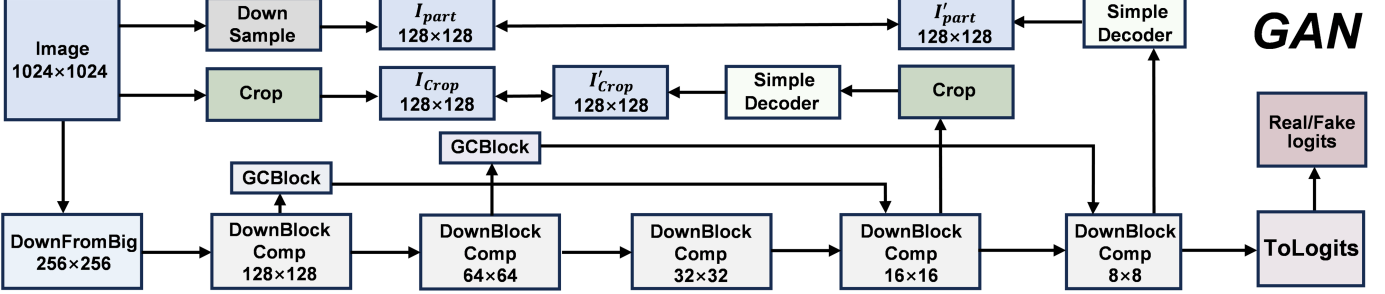


Fig. 2. The structure of our GAN. **a** is the generator network  $G$ , **b** is the discriminator network  $D$ .

BlockCompRes upsampling strategy. The architecture consists of two main networks: the generator  $G$  and discriminator  $D$ , as shown in Fig. 2. Both networks employ global context mechanisms to improve image quality and stabilize training, particularly in scenarios involving high-resolution images.

1) *Generator Network*:  $G$  comprises four main components: InitLayer, UpBlockCompRes, UpBlock, and GCBlock, as illustrated in Fig. 2 **a**. These components work together to progressively generate high-resolution images from random noise, handling both upsampling and feature refinement to ensure image quality and detail preservation.

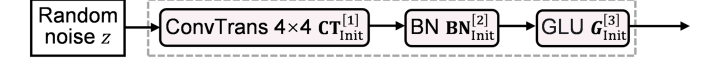
In the proposed GAN model,  $G$  takes a random noise vector  $z$  with dimensions  $d \times 1 \times 1$  as input. The vector  $z$  follows the distribution  $z \sim p_z(z)$ , where  $p_z$  denotes the probability distribution of the noise.

$G$  in GAN begins with InitLayer, which transforms  $z$  into a low-resolution feature map, forming the basis for subsequent upsampling, as shown in Fig. 3 **a**. A  $4 \times 4$  ConvTranspose kernel is applied in the transposed convolution layer to expand the input. The output,  $\text{CT}_{\text{Init}}^{[1]}$ , is passed through a batch normalization (BN) layer. The Gated Linear Unit (GLU) then splits the input into two parts and controls it via a sigmoid function. The corresponding mathematical expressions are:

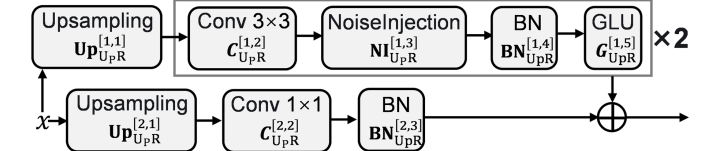
$$\begin{aligned} \text{CT}_{\text{Init}}^{[1]} &= \text{ConvTrans}(z, \mathbf{W}, \mathbf{b}), \\ \text{BN}_{\text{Init}}^{[2]} &= \text{BN}(\text{CT}_{\text{Init}}^{[1]}) = \gamma \frac{\text{CT}_{\text{Init}}^{[1]} - \mu_{\text{batch}}}{\sqrt{\sigma_{\text{batch}}^2 + \epsilon}} + \beta, \\ \mathbf{G}_{\text{Init}}^{[3]} &= \text{GLU}(\text{BN}_{\text{Init}}^{[2]}) = \text{BN}_{\text{Init}}^{[2]A} \odot \sigma(\text{BN}_{\text{Init}}^{[2]B}), \end{aligned} \quad (1)$$

where  $z$  represents the input noise vector,  $\mathbf{W}$  and  $\mathbf{b}$  denote the weights and biases of the transposed convolution.  $\gamma$  and  $\beta$  are the learnable scaling and shifting parameters of the BN layer.  $\mu_{\text{batch}}$  and  $\sigma_{\text{batch}}^2$  refer to the mini-batch mean and variance, while  $\epsilon$  prevents division by zero.  $\sigma$  represents the sigmoid

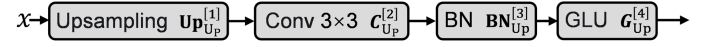
### a. InitLayer



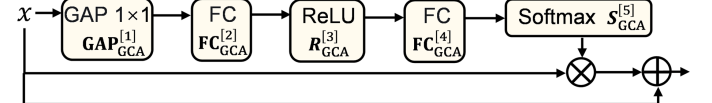
### b. UpBlockCompRes



### c. UpBlock



### d. GCA



### e. GCBlock

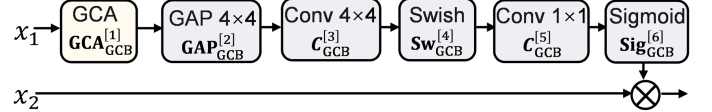


Fig. 3. The structure of the key modules in the  $G$ , where **a** is InitLayer, **b** is UpBlockCompRes, **c** is UpBlock, **d** is GCA, and **e** is GCBlock.

activation function, with  $\text{BN}_{\text{Init}}^{[2]A}$  and  $\text{BN}_{\text{Init}}^{[2]B}$  being the two input components in the GLU.

A key component of  $G$  is UpBlockCompRes, which handles both upsampling and feature compression, as shown in Fig. 3 **b**. In this module, the input feature map  $x$  undergoes upsampling to increase spatial resolution, followed by a sequence of convolutional layers, noise injection, BN, and GLU. After upsampling, a  $3 \times 3$  convolution is applied, and noise is injected into the result. The feature map is then normalized by a BN layer, split into two parts, and processed by the GLU, with a sigmoid function controlling the gating. This process of

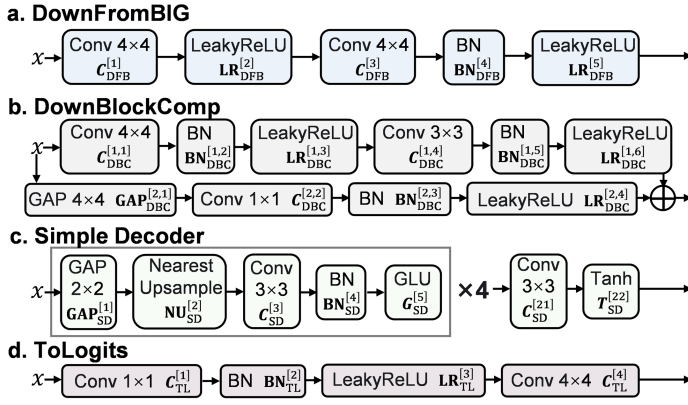


Fig. 4. The structure of the key modules in the  $D$ , where **a** is DownFromBIG, **b** is DownBlockComp, **c** is SimpleDecoder, and **d** is ToLogits.

convolution, noise injection, BN, and GLU is repeated twice to refine the feature representation. Meanwhile, a residual branch upsamples the input feature map, applies a  $1 \times 1$  convolution, and normalizes it via BN to create a residual feature map. Finally, the main and residual feature maps are combined through element-wise addition, forming the block's output. The equations below describe the process:

$$\begin{aligned}
 \text{Up}_{\text{UpR}}^{[1,1]} &= \text{Upsample}(x), \text{C}_{\text{UpR}}^{[1,2]} = \text{Conv}(\text{Up}_{\text{UpR}}^{[1,1]}, \mathbf{W}, \mathbf{b}), \\
 \text{NI}_{\text{UpR}}^{[1,3]} &= \text{C}_{\text{UpR}}^{[1,2]} + \mathbf{n}, \text{BN}_{\text{UpR}}^{[1,4]} = \text{BN}(\text{NI}_{\text{UpR}}^{[1,3]}), \\
 \text{G}_{\text{UpR}}^{[1,5]} &= \text{GLU}(\text{BN}_{\text{UpR}}^{[1,4]}), \text{Up}_{\text{UpR}}^{[2,2]} = \text{Upsample}(x), \\
 \text{C}_{\text{UpR}}^{[2,2]} &= \text{Conv}(\text{Up}_{\text{UpR}}^{[2,2]}, \mathbf{W}, \mathbf{b}), \text{BN}_{\text{UpR}}^{[2,3]} = \text{BN}(\text{C}_{\text{UpR}}^{[2,2]}), \\
 y_{\text{UpR}} &= \text{G}_{\text{UpR}}^{[1,9]} + \text{BN}_{\text{UpR}}^{[2,3]}, \quad (2)
 \end{aligned}$$

where  $\mathbf{n}$  denotes the noise vector injected into the feature map. The superscript of each symbol, such as  $[1, 2]$ , indicates the second feature map in the first branch. UpBlock is designed to increase the resolution of feature maps, as shown in Fig. 3 c. The mathematical expressions are similar to those of the previous module, so the detailed derivation is omitted.

GCBLOCK is a key innovation that enhances global feature perception through GCA, as illustrated in Fig. 3 d. Primarily composed of GCA, GCBLOCK ensures that  $G$  considers the overall image structure during generation, rather than focusing solely on local details, as illustrated in Fig. 3 e. The process is described by the following equations:

$$\begin{aligned}
 \text{GAP}_{\text{GCA}}^{[1]} &= \text{GAP}(x) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x(i, j), \\
 \text{FC}_{\text{GCA}}^{[2]} &= \mathbf{W} \cdot \text{GAP}_{\text{GCA}}^{[1]} + \mathbf{b}, \\
 \text{R}_{\text{GCA}}^{[3]} &= \text{ReLU}(\text{FC}_{\text{GCA}}^{[2]}) = \max(0, \text{FC}_{\text{GCA}}^{[2]}), \\
 \text{FC}_{\text{GCA}}^{[4]} &= \mathbf{W} \cdot \text{R}_{\text{GCA}}^{[3]} + \mathbf{b}, \\
 \text{S}_{\text{GCA}}^{[5]}(i) &= \text{Softmax}(\text{FC}_{\text{GCA}}^{[4]}) = \frac{\exp(\text{FC}_{\text{GCA}}^{[4]}(i))}{\sum_j \exp(\text{FC}_{\text{GCA}}^{[4]}(j))}, \\
 \text{ATT}_{\text{GCA}} &= x \odot \text{S}_{\text{GCA}}^{[5]}, y_{\text{GCA}} = x + \text{ATT}_{\text{GCA}}, \quad (3)
 \end{aligned}$$

where  $x$  represents the input feature map with spatial dimensions  $H \times W$ , and  $\text{GAP}_{\text{GCA}}^{[1]}$  is the result of applying global average pooling (GAP) to  $x$ .  $\mathbf{W}$  and  $\mathbf{b}$  denote the weights and biases of the FC layers.  $\text{ATT}_{\text{GCA}}$  is the feature map obtained after applying attention weights to the original input  $x$ , and  $y_{\text{GCA}}$  is the residual output, combining the original input with the attention-weighted feature map  $x_{\text{att}}$ . Additionally, GCBLOCK uses  $1 \times 1$  and  $4 \times 4$  convolutions, along with Swish and Sigmoid functions, to further refine the feature maps, balancing global consistency and local detail in the generated images. The following equations describe the process:

$$\begin{aligned}
 \text{GCA}_{\text{GCB}}^{[1]} &= \text{GCA}(x_1), \text{GAP}_{\text{GCB}}^{[2]} = \text{GCA}(\text{GAP}_{\text{GCB}}^{[1]}), \\
 \text{C}_{\text{GCB}}^{[3]} &= \text{Conv}(\text{GAP}_{\text{GCB}}^{[2]}, \mathbf{W}, \mathbf{b}), \\
 \text{Sw}_{\text{GCB}}^{[4]} &= \text{C}_{\text{GCB}}^{[3]} \cdot \sigma(\text{C}_{\text{GCB}}^{[3]}) = \text{C}_{\text{GCB}}^{[3]} \cdot \frac{1}{1 + \exp(-\text{C}_{\text{GCB}}^{[3]})}, \\
 \text{C}_{\text{GCB}}^{[5]} &= \text{Conv}(\text{Sw}_{\text{GCB}}^{[4]}, \mathbf{W}, \mathbf{b}), \text{Sig}_{\text{GCB}}^{[6]} = \sigma(\text{C}_{\text{GCB}}^{[5]}), \\
 y_{\text{GCB}} &= x_1 + (x_2 \odot \text{Sig}_{\text{GCB}}^{[6]}), \quad (4)
 \end{aligned}$$

where  $x_1$  is a feature map with dimensions  $512 \times 8 \times 8$ , representing a lower-resolution, high-channel map, while  $x_2$  is a feature map with dimensions  $64 \times 128 \times 128$ , representing a higher-resolution, low-channel map. These feature maps are combined through the subsequent operations in the GCBLOCK.

2) *Discriminator Network*:  $D$  consists of several modules, including DownFromBIG, DownBlockComp, GCBLOCK, SimpleDecoder, and ToLogits, as shown in Fig. 2 b. DownFromBIG reduces the high-resolution input image to a lower-resolution feature map through convolution and downsampling, effectively lowering computational complexity while preserving important low-level features, as shown in Fig. 4 a. DownBlockComp, a crucial component of  $D$ , is responsible for further feature extraction and progressive resolution reduction of the feature map, ensuring detailed information is retained as the resolution decreases, as shown in Fig. 4 b. Another essential module in  $D$  is SimpleDecoder, which reconstructs intermediate feature maps during discrimination, as shown in Fig. 4 c. SimpleDecoder combines GAP, convolution layers, nearest-neighbor upsampling, and batch normalization, progressively enlarging the feature maps. This reconstruction assists  $G$  in refining image details and helps  $D$  extract more comprehensive features. Through this capability,  $D$  not only distinguishes between real and generated images but also performs feature reconstruction, offering stronger feedback and improving adversarial network training. Finally, ToLogits converts the processed feature maps into logits for classification output, as shown in Fig. 4 d. Since the structures of DownFromBIG, DownBlockComp, SimpleDecoder, and ToLogits are similar to previously introduced components, their mathematical formulations are not reiterated.

3) *Loss Function*: The loss function comprises two main components: adversarial loss and reconstruction loss, both of which are crucial for training  $G$  and  $D$ . The reconstruction loss regularizes  $D$  by treating it as an encoder paired with small decoders. This forces  $D$  to extract meaningful features that the



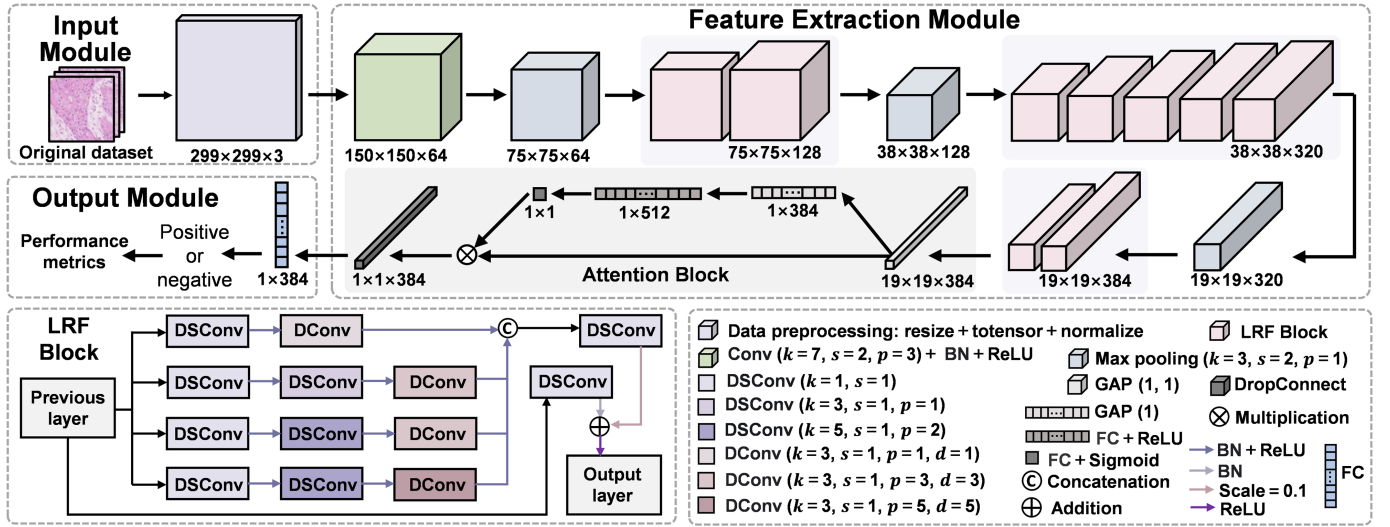


Fig. 5. The architecture of the LRF-CNN model.  $k$  is the convolution kernel size,  $s$  is the stride,  $p$  is the padding, and  $d$  is the dilated rate.

decoders can use to reconstruct input images. Applied only to real images, this loss encourages  $D$  not only to distinguish between real and generated images but also to reconstruct parts of real images. The reconstruction loss is defined as:

$$\mathcal{L}_{\text{recons}} = \mathbb{E}_{\mathbf{f} \sim D_{\text{encode}}(x), x \sim I_{\text{real}}} [\|\mathcal{G}(\mathbf{f}) - \mathcal{T}(x)\|], \quad (5)$$

where  $\mathbf{f}$  represents intermediate feature maps generated by  $D$  from the real image  $x$ ,  $\mathcal{G}$  denotes the decoder applied to the feature maps, and  $\mathcal{T}(x)$  represents real image processing, including transformations like random cropping or downsampling of the real image  $x$ . The L1 norm  $\|\cdot\|$  measures the difference between the decoder's reconstructed output and the processed real image portion  $\mathcal{T}(x)$ . The adversarial loss drives the competition between  $G$  and  $D$  and is defined as:

$$\begin{aligned} \mathcal{L}_D &= -\mathbb{E}_{x \sim I_{\text{real}}} [\min(0, -1 + D(x))] \\ &\quad -\mathbb{E}_{\hat{x} \sim G(z)} [\min(0, -1 - D(\hat{x}))] + \mathcal{L}_{\text{recons}}, \\ \mathcal{L}_G &= -\mathbb{E}_{z \sim \mathcal{N}} [D(G(z))], \end{aligned} \quad (6)$$

where  $D(x)$  is the output of  $D$  for real images, and  $D(\hat{x})$  is its output for generated images  $\hat{x}$  from  $G$ . The hinge loss ensures that  $D$  confidently classifies real images as real and generated images as fake. The term  $\min(0, -1 + D(x))$  penalizes  $D$  when it lacks confidence in classifying real images, while  $\min(0, -1 - D(\hat{x}))$  penalizes misclassification of generated images. The inclusion of  $\mathcal{L}_{\text{recons}}$  ensures that  $D$  also performs feature reconstruction, enriching extracted features. The term  $D(G(z))$  represents  $D$ 's output for images generated by  $G$  from  $z$ , with  $G$  aiming to minimize  $\mathcal{L}_G$ , encouraging  $D$  to classify generated images as real.

### B. LRF-CNN Architecture

The LRF-CNN model is designed to classify histopathological images in this study, as depicted in Fig. 5. The architecture is composed of three key components: Input Module, Feature Extraction Module, and Output Module.

Input Module prepares the original histopathological images for processing. The images are resized to a fixed dimension of  $299 \times 299 \times 3$ , converted into tensors, and normalized to ensure consistency and enhance quality. Feature Extraction Module includes max pooling, LRF Blocks, and an Attention Block, which are responsible for feature extraction. LRF Block enhances feature extraction by combining multi-scale and dilated convolutions. It consists of multiple branches, each performing convolutions with different kernel sizes and dilation rates. This multi-branch design captures both local and global features, expanding the receptive field without significantly increasing computational complexity. Attention Block refines the features extracted by the LRF Block, emphasizing the most relevant information for classification. Output Module includes a fully connected layer and a Softmax function. It computes the probability distribution across multiple classes, providing the final classification results and performance metrics.

## IV. EXPERIMENTS AND RESULTS

### A. Experiment Details

1) *Datasets*: This study utilizes two histopathological image datasets from different cancers, with classification performed on each dataset individually. The esophageal cancer dataset [8] contains 3744 positive Programmed Death Ligand-1 (PD-L1) state images and 2591 negative PD-L1 state images, all in the “.png” format with a resolution of  $290 \times 320$  pixels. For breast cancer, images from the BreakHis dataset [23] are selected at  $200\times$  magnification, comprising 623 benign and 1390 malignant breast tumor images, with a resolution of  $700 \times 460$  pixels, also in the “.png” format. Both datasets exhibit significant class imbalance, with certain categories having many more samples than others, as shown in Fig. 6.

2) *Experimental Settings*: This study involves histopathological image generation and classification, with detailed experimental settings outlined as follows. For histopathological

TABLE I

FID, IS, PSNR, AND SSIM COMPARISONS ON THE ESOPHAGEAL CANCER DATASET USING FASTGAN AND OUR GAN AT  $1024 \times 1024$  RESOLUTION.

| Training iteration        | 40K          |             |              |             | 60K          |             |              |             | 80K          |             |              |             |
|---------------------------|--------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|
| Criterion                 | FID          | IS          | PSNR         | SSIM        | FID          | IS          | PSNR         | SSIM        | FID          | IS          | PSNR         | SSIM        |
| Baseline (FastGAN)        | 99.64        | 1.69        | 10.73        | 0.25        | 88.98        | 1.65        | 11.14        | <b>0.26</b> | 99.58        | 1.70        | 10.33        | 0.26        |
| Baseline + GCBBlock       | 107.07       | 1.72        | <b>11.12</b> | 0.23        | 82.31        | 1.85        | 11.13        | 0.25        | 75.04        | 1.72        | 10.56        | 0.25        |
| Baseline + UpBlockCompRes | 95.67        | 1.78        | 10.77        | 0.25        | 80.53        | <b>1.87</b> | 10.57        | <b>0.26</b> | 77.25        | 1.79        | 10.61        | 0.25        |
| Our GAN                   | <b>84.21</b> | <b>1.94</b> | 11.02        | <b>0.27</b> | <b>72.34</b> | 1.73        | <b>11.21</b> | <b>0.26</b> | <b>67.12</b> | <b>1.83</b> | <b>11.05</b> | <b>0.27</b> |

TABLE II

FID, IS, PSNR, AND SSIM COMPARISONS ON THE BREAKHIS DATASET USING FASTGAN AND OUR GAN AT  $1024 \times 1024$  RESOLUTION.

| Training iteration        | 40K           |             |             |             | 60K           |             |             |             | 80K           |             |             |             |
|---------------------------|---------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|---------------|-------------|-------------|-------------|
| Criterion                 | FID           | IS          | PSNR        | SSIM        | FID           | IS          | PSNR        | SSIM        | FID           | IS          | PSNR        | SSIM        |
| Baseline (FastGAN)        | 162.33        | 2.60        | 8.39        | 0.28        | 151.89        | 2.63        | 8.87        | 0.28        | 142.68        | 2.52        | 8.60        | <b>0.30</b> |
| Baseline + GCBBlock       | 155.91        | 2.50        | 8.29        | 0.27        | 145.85        | 2.51        | <b>9.26</b> | <b>0.31</b> | 143.18        | 2.51        | 8.73        | 0.29        |
| Baseline + UpBlockCompRes | 155.91        | 2.49        | 8.33        | 0.28        | 145.53        | <b>2.55</b> | 8.53        | 0.29        | 139.29        | <b>2.59</b> | 8.58        | 0.28        |
| Our GAN                   | <b>154.58</b> | <b>2.62</b> | <b>8.42</b> | <b>0.29</b> | <b>144.33</b> | 2.45        | 8.22        | 0.29        | <b>138.31</b> | 2.52        | <b>9.15</b> | <b>0.30</b> |

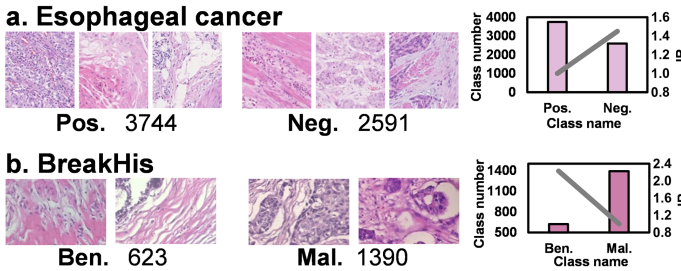


Fig. 6. Examples of histopathological image datasets. **a** are examples of esophageal cancer. **b** are examples of BreakHis in breast cancer. The words in the figure (e.g., Pos.) represent the name of a category, while the numbers (e.g., 3744) represent the number of a category. The chart on the right of each dataset is about the number and imbalance rate (IR) of each category.

image generation, the experiments focus on our GAN and FastGAN [22], conducted on a local machine equipped with 32 GB RAM and an 8 GB NVIDIA Quadro RTX 4000 GPU, using Python 3.8.0 and PyTorch 2.0.0+cu117. The setup includes a batch size of 2, an image resolution of  $1024 \times 1024$ , a generated image size of  $512 \times 512$ , and 80,000 training iterations. For histopathological image classification, the model is trained for 100 epochs with a batch size of 8, using the same machine configuration. The dataset is split into 60% training, 20% validation, and 20% testing. The Adam optimizer is employed with a learning rate of 0.0001 and weight decay of  $1e-5$ , while StepLR halves the learning rate every 20 epochs. The cross-entropy loss function is used to train the LRF-CNN model.

3) *Performance Metrics*: The performance evaluation in this study covers both histopathological image generation and classification tasks, with specific metrics for each. For image generation, Frechet Inception Distance (FID), Inception Score (IS), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM) are used to assess the quality of the generated images. In the classification task, due to class imbalance in the dataset, several unbiased metrics are employed,

including Precision (P), Recall (R), F1 Score (F1), Geometric Mean (G-mean), Specificity (S), and Matthews Correlation Coefficient (MCC). Although Accuracy (Acc) is also reported, it is acknowledged to be biased in imbalanced scenarios and is primarily included for comparison purposes.

### B. GAN-based Generation of Histopathological Images

In the image generation task, the esophageal cancer and BreakHis datasets serve as benchmarks to compare the performance of FastGAN and our GAN. As shown in Table I, FastGAN exhibits higher FID values across all training iterations (e.g., 99.64 at 40K) and lower IS values, indicating poorer image quality and diversity. In contrast, the proposed GAN significantly reduces FID (e.g., 84.21 at 40K) and improves IS, indicating the generated images are closer to real data in terms of visual quality and diversity. The proposed GAN also outperforms FastGAN in PSNR and SSIM, especially at 80K iterations, where PSNR reaches 11.05 and SSIM 0.27, showing that the generated images are structurally closer to the real data. Similarly, as shown in Table II, FastGAN reports higher FID values on the BreakHis dataset (e.g., 162.33 at 40K), and its IS remains relatively low, reflecting a significant deviation from real data. In contrast, the proposed GAN reduces FID (e.g., 154.58 at 40K) and improves IS and PSNR, with IS reaching 2.62 and PSNR 8.42 at 40K. As illustrated in Fig. 7, real histopathological images are compared with those generated by FastGAN and our GAN across three categories. Our GAN consistently produces images more similar to real data, particularly in terms of color and texture details.

### C. LRF-CNN Classification Results with Data Augmentation

In the esophageal cancer experiments, 25 images per category generated by FastGAN and the proposed GAN are added to the original dataset, followed by classification using LRF-CNN, as shown in Table III. With 50 images from the proposed GAN, classification performance improves notably, achieving

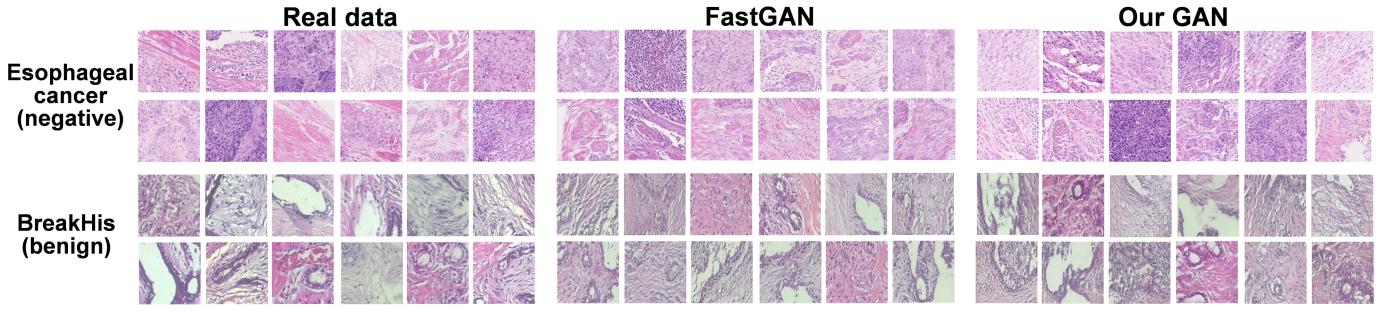


Fig. 7. The comparison between real data images and images generated by FastGAN and our GAN. The first column displays the real data images, followed by images generated by FastGAN in the second column, and our GAN in the final column. The first row represents the results of esophageal cancer, and the second row represents the results of BreakHis.

TABLE III  
PERFORMANCE OF LRF-CNN ON THE ESOPHAGEAL CANCER TEST SET USING DATA AUGMENTATION BASED ON FASTGAN AND OUR GAN.

| Models               | Data                          | Acc          | P            | R            | F1           | G-mean       | S            | MCC          |
|----------------------|-------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| No data augmentation | Original dataset              | 79.01        | 81.42        | 82.74        | 82.08        | 78.16        | 73.82        | 56.76        |
| FastGAN              | Original dataset + 25 images  | 77.75        | 80.85        | 81.60        | 81.22        | 76.77        | 72.22        | 53.94        |
|                      | Original dataset + 50 images  | 81.05        | <b>85.90</b> | 83.08        | 84.47        | 80.36        | <b>77.73</b> | 60.24        |
|                      | Original dataset + 75 images  | 78.71        | 80.66        | 82.95        | 81.79        | 77.78        | 72.93        | 56.20        |
|                      | Original dataset + 100 images | 78.60        | 77.58        | 83.84        | 75.51        | 79.46        | 77.90        | 80.37        |
| Our GAN              | Original dataset + 25 images  | 80.74        | 83.44        | 84.00        | 83.72        | 79.93        | 76.05        | 60.14        |
|                      | Original dataset + 50 images  | <b>81.83</b> | 85.18        | <b>85.61</b> | <b>85.39</b> | <b>80.48</b> | 75.67        | <b>61.38</b> |
|                      | Original dataset + 75 images  | 79.41        | 80.25        | 85.25        | 82.68        | 78.05        | 71.45        | 57.51        |
|                      | Original dataset + 100 images | 78.75        | 82.69        | 80.08        | 81.18        | 78.30        | 76.96        | 57.10        |

TABLE IV  
PERFORMANCE OF LRF-CNN ON THE BREAKHIS TEST SET USING DATA AUGMENTATION BASED ON FASTGAN AND OUR GAN.

| Models               | Data                         | Acc          | P            | R            | F1           | G-mean       | S            | MCC          |
|----------------------|------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| No data augmentation | Original dataset             | 98.27        | 98.26        | 99.30        | 98.77        | 97.55        | 95.83        | 95.84        |
| FastGAN              | Original dataset + 10 images | 97.29        | 98.25        | 97.90        | 98.07        | 96.86        | 95.83        | 93.51        |
|                      | Original dataset + 20 images | 97.30        | 97.58        | 98.60        | 98.09        | 96.41        | 94.26        | 93.54        |
|                      | Original dataset + 30 images | 97.56        | 96.98        | <b>99.66</b> | 98.30        | 96.01        | 92.50        | 94.10        |
|                      | Original dataset + 40 images | 96.84        | 97.34        | 97.71        | 97.52        | 96.51        | 95.33        | 93.18        |
| Our GAN              | Original dataset + 10 images | 97.78        | 98.26        | 98.60        | 98.43        | 97.21        | 95.83        | 94.67        |
|                      | Original dataset + 20 images | 98.28        | 98.95        | 98.60        | 98.77        | 98.07        | 97.54        | 95.92        |
|                      | Original dataset + 30 images | <b>98.29</b> | <b>99.65</b> | 97.93        | <b>98.78</b> | <b>98.55</b> | <b>98.17</b> | <b>95.97</b> |
|                      | Original dataset + 40 images | 96.60        | 95.93        | 98.85        | 97.37        | 95.71        | 92.67        | 92.66        |

Acc of 81.83%, R of 85.61%, F1 rising to 85.39%, G-mean at 80.48%, and MCC at 61.38%. In contrast, FastGAN-generated images lead to Acc of 81.05% with smaller gains in R (83.08%) and G-mean (80.36%), and MCC (60.24%). These results indicate that the proposed GAN generates images that better enrich the dataset and capture essential features, outperforming FastGAN in metrics such as R and F1.

In the BreakHis dataset experiments, 10 images per category are added to the original dataset, as shown in Table IV. When 30 images from the proposed GAN are used, classification performance reaches Acc of 98.29%, P of 99.65%, F1 of 98.78%, G-mean of 98.55%, and S of 98.17%. FastGAN-generated images result in Acc of 97.56%, with P at 96.98%, F1 at 98.30%, and MCC at 94.10%. While FastGAN shows some improvement in R, its overall performance is still inferior

to the proposed GAN, demonstrating that the proposed method consistently performs better and surpasses FastGAN.

## V. DISCUSSION

As shown in Fig. 8, with the images generated by the proposed GAN, the model improves its accuracy in predicting positive cases within the esophageal cancer dataset, reducing false negatives from 127 (without augmentation) to 114, while FastGAN results in 134 false negatives. For negative cases, false positives drop from 139 to 118 using the proposed GAN. This highlights its effectiveness in minimizing false negatives compared to FastGAN, improving the distinction between positive and negative cases. In the BreakHis dataset, false negatives for benign cases decrease from 5 to 1 with the proposed GAN, while FastGAN results in 9 false negatives. However, for malignant cases, false positives increase from



2 to 6 with the proposed GAN, while FastGAN results in only 1. Although our GAN reduces false negatives for benign cases more effectively, it is less successful in controlling false positives for malignant cases compared to FastGAN.

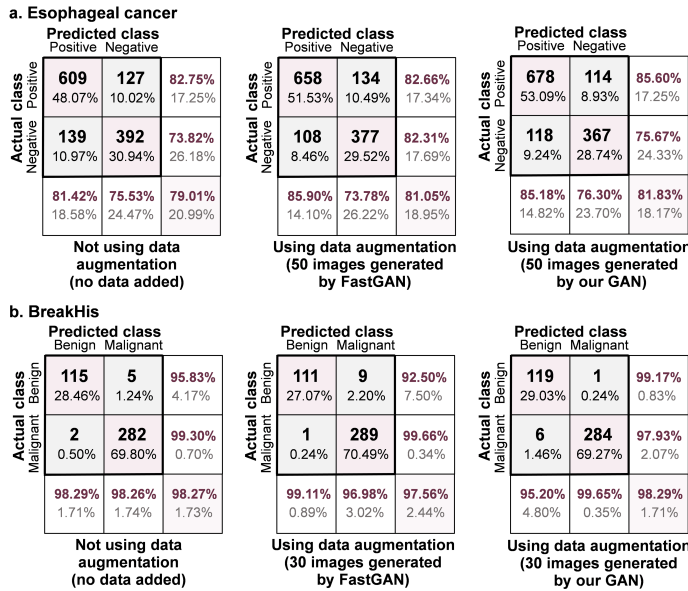


Fig. 8. Confusion matrices for classification experiments with and without using data augmentation. Red and gray numbers are the percentage of correct and incorrect cases, respectively. **a** is the confusion matrix of esophageal cancer, and **b** is the confusion matrix of BreakHis.

## VI. CONCLUSION AND FUTURE WORK

This study introduces a GAN-based data augmentation method to tackle class imbalance in histopathological image classification. By generating high-quality images for minority classes, the GAN increases data diversity and reduces bias. Our method also has a potential to be used in related biomedical image analysis fields, such as [24]–[28].

However, false positives, especially for malignant cases, remain a challenge. This suggests further improvements are needed in generating samples for complex categories. Future work will focus on optimizing the GAN to reduce false positives and exploring hybrid models or new loss functions to enhance performance in multi-class classification tasks.

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